PW 01: Data Preparation

Objective:

Understand the importance of **data preparation** — including **data cleaning, transformation, and organization** — to ensure data quality and reliability before analysis.

1- Introduction to Data Science with Pandas

- **Objective:** Understand the basics of data science using Pandas by learning how to import, explore, and summarize datasets. Develop familiarity with core DataFrame operations such as selection, filtering, and basic statistics.

In [105]:
```
import pandas as pd
```

In [108]:
```
# load data
df = pd.read_csv ( 'cars.csv' )
#df = pd.read_json ( 'cars.json' )
#df = pd.read_excel ( 'cars.xlsx' )
```

In [107]:
```
type(df)
```

Out[107]:
```
pandas.core.frame.DataFrame
```

- **DataFrame Overview**

In [111]:
```
df.head(7)
```

Out[111]:

	Unnamed: 0	car_name	brand	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type	mileage	engine	max_power	seats	price
0	0	Maruti Alto	Maruti	Alto	9	120000	Individual	Petrol	Manual	19.70	796	46.30	5	120000
1	1	Hyundai Grand	Hyundai	Grand	5	20000	Individual	Petrol	Manual	18.90	1197	82.00	5	550000
2	2	Hyundai i20	Hyundai	i20	11	60000	Individual	Petrol	Manual	17.00	1197	80.00	5	215000
3	3	Maruti Alto	Maruti	Alto	9	37000	Individual	Petrol	Manual	20.92	998	67.10	5	226000
4	4	Ford Ecosport	Ford	Ecosport	6	30000	Dealer	Diesel	Manual	22.77	1498	98.59	5	570000
5	5	Maruti Wagon R	Maruti	Wagon R	8	35000	Individual	Petrol	Manual	18.90	998	67.10	5	350000
6	6	Hyundai i10	Hyundai	i10	8	40000	Dealer	Petrol	Manual	20.36	1197	78.90	5	315000

In [113]:
```
df.tail(3)
```

Out[113]:

	Unnamed: 0	car_name	brand	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type	mileage	engine	max_power	seats	price
15408	19541	Skoda Rapid	Skoda	Rapid	6	67000	Dealer	Diesel	Manual	21.14	1498	103.52	5	425000
15409	19542	Mahindra XUV500	Mahindra	XUV500	5	3800000	Dealer	Diesel	Manual	16.00	2179	140.00	7	1225000
15410	19543	Honda City	Honda	City	2	13000	Dealer	Petrol	Automatic	18.00	1497	117.60	5	1200000

In [116]:
```
df.sample(3) # get 3 rows randomly
```

Out[116]:

	Unnamed: 0	car_name	brand	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type	mileage	engine	max_power	seats	price
5392	6842	Hyundai i10	Hyundai	i10	11	90000	Dealer	Petrol	Manual	20.36	1197	78.90	5	275000
10427	13292	Maruti Vitara	Maruti	Vitara	4	75000	Dealer	Diesel	Manual	24.30	1248	88.50	5	775000
10546	13444	Skoda Superb	Skoda	Superb	7	84000	Dealer	Diesel	Automatic	17.20	1968	138.08	5	1040000

In [118]:
```
df.columns # get list of columns
```

Out[118]:
```
['Unnamed: 0',
 'car_name',
 'brand',
 'model',
 'vehicle_age',
 'km_driven',
 'seller_type',
 'fuel_type',
 'transmission_type',
 'mileage',
 'engine',
 'max_power',
 'seats',
 'price']
```

In [121]:
```
df.dtypes # get data types of columns
```

Out[121]:
```
Unnamed: 0             int64
car_name              object
brand                 object
model                 object
vehicle_age            int64
km_driven              int64
seller_type           object
fuel_type             object
transmission_type     object
mileage
                     float64
engine                 int64
max_power            float64
seats                  int64
price                  int64
dtype: object
```

In [122]:
```
df.info() # get summary of dataframe
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15411 entries, 0 to 15410
Data columns (total 14 columns):
 #    Column             Non-Null Count Dtype
---  ------             -------------- -----
 0    Unnamed: 0         15411 non-null int64
 1    car_name           15411 non-null object
 2    brand              15411 non-null object
 3    model              15411 non-null object
 4    vehicle_age        15411 non-null int64
 5    km_driven          15411 non-null int64
 6    seller_type        15411 non-null object
 7    fuel_type          15411 non-null object
 8    transmission_type 15411 non-null object
 9    mileage            15411 non-null float64
 10 engine               15411 non-null int64
 11 max_power            15411 non-null float64
 12 seats                15411 non-null int64
 13 price                15411 non-null int64
dtypes: float64(2), int64(6), object(6)
memory usage: 1.6+ MB
```

In [ ]:

In [ ]:

In [ ]:

- **Basic Statistics**

In [124]:
```
df.describe().round(2) # get numerical statistical sumary
```

Out[124]:

	Unnamed: 0	vehicle_age	km_driven	mileage	engine	max_power	seats	price
count	15411.00	15411.00	15411.00	15411.00	15411.00	15411.00	15411.00	15411.00
mean	9811.86	6.04	55616.48	19.70	1486.06	100.59	5.33	774971.12
std	5643.42	3.01	51618.55	4.17	521.11	42.97	0.81	894128.36
min	0.00	0.00	100.00	4.00	793.00	38.40	0.00	40000.00
25%	4906.50	4.00	30000.00	17.00	1197.00	74.00	5.00	385000.00
50%	9872.00	6.00	50000.00	19.67	1248.00	88.50	5.00	556000.00
75%	14668.50	8.00	70000.00	22.70	1582.00	117.30	5.00	825000.00
max	19543.00	29.00	3800000.00	33.54	6592.00	626.00	9.00	39500000.00

In [125]:
```
df["brand"].value_counts() # frequency of values
```

Out[125]:
```
brand
Maruti            4992
Hyundai           2982
Honda             1485
Mahindra          1011
Toyota             793
Ford               790
Volkswagen         620
Renault            536
BMW                439
Tata
                  430
Mercedes-Benz      337
Skoda              334
Audi               192
Datsun             170
Jaguar              59
Land Rover          51
Jeep                41
Kia                 32
Porsche             21
Volvo               20
MG                  19
Mini                17
Nissan              11
Lexus               10
Isuzu                8
Bentley              3
Maserati             2
ISUZU                2
Ferrari              1
Mercedes-AMG         1
Rolls-Royce          1
Force                1
Name: count, dtype: int64
```

In [126]:
```
df["brand"].unique() # get list of possible values
```

Out[126]:
```
array(['Maruti', 'Hyundai', 'Ford', 'Renault', 'Mini', 'Mercedes-Benz',
       'Toyota', 'Volkswagen', 'Honda', 'Mahindra', 'Datsun', 'Tata',
       'Kia', 'BMW', 'Audi', 'Land Rover', 'Jaguar', 'MG', 'Isuzu',
       'Porsche', 'Skoda', 'Volvo', 'Lexus', 'Jeep', 'Maserati',
       'Bentley', 'Nissan', 'ISUZU', 'Ferrari', 'Mercedes-AMG',
       'Rolls-Royce', 'Force'], dtype=object)
```

In [127]:
```
df["brand"].nunique() # get number of possible values
```

Out[127]:
```
32
```

In [ ]:
```

```

In [ ]:
```

```

- **Column/Row Selection**

In [128]:
```
df["brand"]
```

Out[128]:
```
0         Maruti
1        Hyundai
2        Hyundai
3         Maruti
4           Ford
          ...
15406    Hyundai
15407     Maruti
15408      Skoda
15409    Mahindra
15410       Honda
Name: brand, Length: 15411, dtype: object
```

In [131]:
```
df[["brand", "car_name"]] # multiple cols
```

Out[131]:

	brand	car_name
0	Maruti	Maruti Alto
1	Hyundai	Hyundai Grand
2	Hyundai	Hyundai i20
3	Maruti	Maruti Alto
4	Ford	Ford Ecosport
...	...	...
15406	Hyundai	Hyundai i10
15407	Maruti	Maruti Ertiga
15408	Skoda	Skoda Rapid
15409	Mahindra	Mahindra XUV500
15410	Honda	Honda City

15411 rows × 2 columns

In [132]:
```
df.iloc[0] # selects row by position
```

Out[132]:
```
Unnamed: 0                    0
car_name             Maruti Alto
brand                     Maruti
model                       Alto
vehicle_age                    9
km_driven                 120000
seller_type           Individual
fuel_type                 Petrol
transmission_type         Manual
mileage                     19.7
engine                       796
max_power                   46.3
seats                          5
price                     120000
Name: 0, dtype: object
```

In [134]:
```
df.iloc[3:5] # selects many rows by position
```

Out[134]:

	Unnamed: 0	car_name	brand	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type	mileage	engine	max_power	seats	price
3	3	Maruti Alto	Maruti	Alto	9	37000	Individual	Petrol	Manual	20.92	998	67.10	5	226000
4	4	Ford Ecosport	Ford	Ecosport	6	30000	Dealer	Diesel	Manual	22.77	1498	98.59	5	570000

In [136]:
```
df2 = df.loc[df["price"] > 10000] # select rows by condition
```

In [140]:
```
df.loc[df["price"] > df["price"].mean(), "car_name"] # select rows by condition
```

Out[140]:
```
8              Hyundai Venue
12               Mini Cooper
15      Mercedes-Benz C-Class
22              Hyundai Creta
29           Toyota Fortuner
                 ...
15397           Toyota Innova
15404           Maruti Ertiga
15407           Maruti Ertiga
15409         Mahindra XUV500
15410             Honda City
Name: car_name, Length: 4365, dtype: object
```

In [141]:
```
print("the average of price of cars = ", df["price"].mean())
```

```
the average of price of cars = 774971.1164103562
```

- **Creating and Deleting Columns**

In [142]:
```
df["km_per_year"] = df["km_driven"] / df["vehicle_age"]
```

In [144]:
```
df
```

Out[144]:

	Unnamed: 0	car_name	brand	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type	mileage	engine	max_power	seats	price	km_per_year
0	0	Maruti Alto	Maruti	Alto	9	120000	Individual	Petrol	Manual	19.70	796	46.30	5	120000	13333.333333
1	1	Hyundai Grand	Hyundai	Grand	5	20000	Individual	Petrol	Manual	18.90	1197	82.00	5	550000	4000.000000
2	2	Hyundai i20	Hyundai	i20	11	60000	Individual	Petrol	Manual	17.00	1197	80.00	5	215000	5454.545455
3	3	Maruti Alto	Maruti	Alto	9	37000	Individual	Petrol	Manual	20.92	998	67.10	5	226000	4111.111111
4	4	Ford Ecosport	Ford	Ecosport	6	30000	Dealer	Diesel	Manual	22.77	1498	98.59	5	570000	5000.000000
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
15406	19537	Hyundai i10	Hyundai	i10	9	10723	Dealer	Petrol	Manual	19.81	1086	68.05	5	250000	1191.444444
15407	19540	Maruti Ertiga	Maruti	Ertiga	2	18000	Dealer	Petrol	Manual	17.50	1373	91.10	7	925000	9000.000000
15408	19541	Skoda Rapid	Skoda	Rapid	6	67000	Dealer	Diesel	Manual	21.14	1498	103.52	5	425000	11166.666667
15409	19542	Mahindra XUV500	Mahindra	XUV500	5	3800000	Dealer	Diesel	Manual	16.00	2179	140.00	7	1225000	760000.000000
15410	19543	Honda City	Honda	City	2	13000	Dealer	Petrol	Automatic	18.00	1497	117.60	5	1200000	6500.000000

15411 rows × 15 columns

In [150]:
```
df.drop(["brand", "car_name"], axis=1) #remove many cols
```

Out[150]:

	Unnamed: 0	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type	mileage	engine	max_power	seats	price
0	0	Alto	9	120000	Individual	Petrol	Manual	19.70	796	46.30	5	120000
1	1	Grand	5	20000	Individual	Petrol	Manual	18.90	1197	82.00	5	550000
2	2	i20	11	60000	Individual	Petrol	Manual	17.00	1197	80.00	5	215000
3	3	Alto	9	37000	Individual	Petrol	Manual	20.92	998	67.10	5	226000
4	4	Ecosport	6	30000	Dealer	Diesel	Manual	22.77	1498	98.59	5	570000
...	...	...	...	...	...	...	...	...	...	...	...	...
15406	19537	i10	9	10723	Dealer	Petrol	Manual	19.81	1086	68.05	5	250000
15407	19540	Ertiga	2	18000	Dealer	Petrol	Manual	17.50	1373	91.10	7	925000
15408	19541	Rapid	6	67000	Dealer	Diesel	Manual	21.14	1498	103.52	5	425000
15409	19542	XUV500	5	3800000	Dealer	Diesel	Manual	16.00	2179	140.00	7	1225000
15410	19543	City	2	13000	Dealer	Petrol	Automatic	18.00	1497	117.60	5	1200000

15411 rows × 12 columns

In [140]:
```
df.columns
```

Out[140]:
```
Index(['Unnamed: 0', 'car_name', 'brand', 'model', 'vehicle_age', 'km_driven',
       'seller_type', 'fuel_type', 'transmission_type', 'mileage', 'engine',
       'max_power', 'seats', 'price'],
      dtype='object')
```

In [ ]:
```

```

In [ ]:
```

```

- **Display the row with the minimum price**

In [152]:
```
df[df["price"] == df["price"].min()]
```

Out[152]:

	Unnamed: 0	car_name	brand	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type	mileage	engine	max_power	seats	price
7607	9731	Maruti Wagon R	Maruti	Wagon R	21	80000	Individual	Petrol	Manual	18.9	998	67.1	5	40000

In [ ]:
```

```

In [ ]:
```

```

In [155]:
```
# sort data by price descending

df.sort_values(by="price", ascending=False, inplace=True)
```

In [156]:
```
df
```

Out[156]:

	Unnamed: 0	car_name	brand	model	vehicle_age	km_driven	seller_type	fuel_type	transmission_type	mileage	engine	max_power	seats	price
3799	4845	Ferrari GTC4Lusso	Ferrari	GTC4Lusso	2	3800	Dealer	Petrol	Automatic	4.00	3855	601.0	4	39500000
10969	13944	Rolls-Royce Ghost	Rolls-Royce	Ghost	4	5000	Individual	Petrol	Automatic	10.20	6592	563.0	4	24200000
1172	1508	Bentley Continental	Bentley	Continental	9	9000	Dealer	Petrol	Automatic	9.50	5998	626.0	4	14500000
9722	12456	Mercedes-Benz S-Class	Mercedes-Benz	S-Class	3	4000	Dealer	Petrol	Automatic	7.81	4663	459.0	4	13000000
9364	12023	Porsche Cayenne	Porsche	Cayenne	4	24000	Dealer	Petrol	Automatic	12.50	3604	440.0	5	11100000
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
7361	9413	Honda City	Honda	City	19	110000	Individual	Petrol	Manual	13.00	1493	100.0	5	50000
3787	4831	Maruti Alto	Maruti	Alto	19	120000	Individual	Petrol	Manual	18.90	1061	47.0	5	50000
12298	15593	Maruti Wagon R	Maruti	Wagon R	17	50000	Individual	Petrol	Manual	18.90	998	67.1	5	50000
13676	17349	Maruti Alto	Maruti	Alto	17	110000	Individual	Petrol	Manual	19.70	796	46.3	5	45000
7607	9731	Maruti Wagon R	Maruti	Wagon R	21	80000	Individual	Petrol	Manual	18.90	998	67.1	5	40000

15411 rows × 14 columns

1- Introduction to Data Science with Pandas

- **Objective:** Understand the basics of data science using Pandas by learning how to import, explore, and summarize datasets. Develop familiarity with core DataFrame operations such as selection, filtering, and basic statistics.

In [1]:
```
import pandas as pd

df = pd.read_excel("sales_data.xlsx")

print(df.head())
print(df.info())
print(df.isnull().sum())
```

```
    country order_value_EUR          cost                 date     category \
0    Sweden          98320.37   77722.25             8/23/2020        Games
1    France          46296.26   40319.41             5/15/2020        Games
2 Portugal         140337.34 115708.14 2020-04-09 00:00:00       Appliances
3    France        203604.46 175344.16               6/26/2019 Electronics
4     Italy        127075.41 110924.13               9/24/2019     Clothing

       customer_name     sales_manager             sales_rep device_type \
0    Konopelski LLC      Maxie Marrow     Tarrah Castelletti      Tablet
1         Wisoky Inc     Othello Bowes Amelina Piscopiello         Tablet
2     Hegmann Group Celine Tumasian            Corene Shirer          PC
3    Kirlin and Sons     Othello Bowes           Crysta Halls      Mobile
4 Labadie and Sons      Piggy Roscrigg         Joshua Prevost          PC

     order_id
0 70-0511466
1 77-3489084
2 65-8218141
3 29-5478106
4 57-6602854
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #    Column            Non-Null Count Dtype
--- ------             -------------- -----
 0    country           1000 non-null    object
 1
 order_value_EUR 995 non-null    float64
 2   cost              999 non-null   float64
 3   date              1000 non-null  object
 4   category          1000 non-null  object
 5   customer_name     1000 non-null  object
 6   sales_manager     1000 non-null  object
 7   sales_rep         1000 non-null  object
 8   device_type       995 non-null   object
 9   order_id          1000 non-null  object
dtypes: float64(2), object(8)
memory usage: 78.3+ KB
None
country            0
order_value_EUR    5
cost               1
date               0
category           0
customer_name      0
sales_manager      0
sales_rep          0
device_type        5
order_id           0
dtype: int64
```

In [ ]:
```
```